

FinBERT Monetary Policy Tone: SBP MPC Minutes and KSE-100 Market Response

A financial NLP and econometrics project using FinBERT to convert State Bank of Pakistan monetary policy communication into a document-level tone index, then testing whether that tone helps explain short-run KSE-100 log returns after MPC announcements.

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Data: SBP MPC minutes, KSE-100 returns, KIBOR rate

Tools: Python, pandas, FinBERT, transformers, nltk, statsmodels, matplotlib

Executive Summary

This project builds a text-based monetary policy tone index from State Bank of Pakistan MPC minutes using FinBERT, a transformer model trained for financial sentiment classification. Each MPC document was split into sentences, each sentence was scored as positive, negative, or neutral, and the sentence-level probabilities were aggregated into one net tone score per document.

The project then links the FinBERT tone index to market data by matching each MPC release date to the next available KSE-100 trading date and adding the KIBOR rate as a monetary/interest-rate control. The econometric part tests whether monetary policy tone explains KSE-100 log returns using lagged ARDL-style regressions, an unrestricted error correction model, residual diagnostics, and a simple OLS benchmark.

The main finding is cautious: the richer UECM specification has stronger overall fit, but the FinBERT tone variables are not individually statistically significant in the final model. The simple contemporaneous OLS model also shows weak explanatory power. This suggests that SBP communication tone alone is not enough to explain short-run KSE-100 returns without a fuller macro-financial setup.

1. Data and Methodology

The text dataset consists of SBP Monetary Policy Committee minutes from 2016 to 2025. Around 70 documents were processed. Each document was cleaned, split into sentences, scored with FinBERT, and converted into a document-level tone score.

The tone construction uses the following logic: positive, negative, and neutral probabilities are estimated for each sentence. The document-level net tone is then calculated as the sum of positive probabilities minus the sum of negative probabilities, divided by the number of sentences.

The resulting tone series was merged with market data. MPC dates were matched to the next available KSE-100 trading day using a forward merge within a seven-day tolerance. KIBOR was added as an interest-rate control.

Component	Description
Text source	SBP MPC minutes, 2016-2025
NLP model	ProsusAI/finbert sentence-level sentiment model
Tone metric	Net tone = average positive probability minus average negative probability
Market outcome	KSE-100 log return on the matched trading date
Control variable	KIBOR rate

2. FinBERT Tone Index

The first output of the project is a time-series index of monetary policy tone. Positive values indicate that the MPC minutes were more positive or optimistic on average; negative values indicate a more negative or cautious tone.

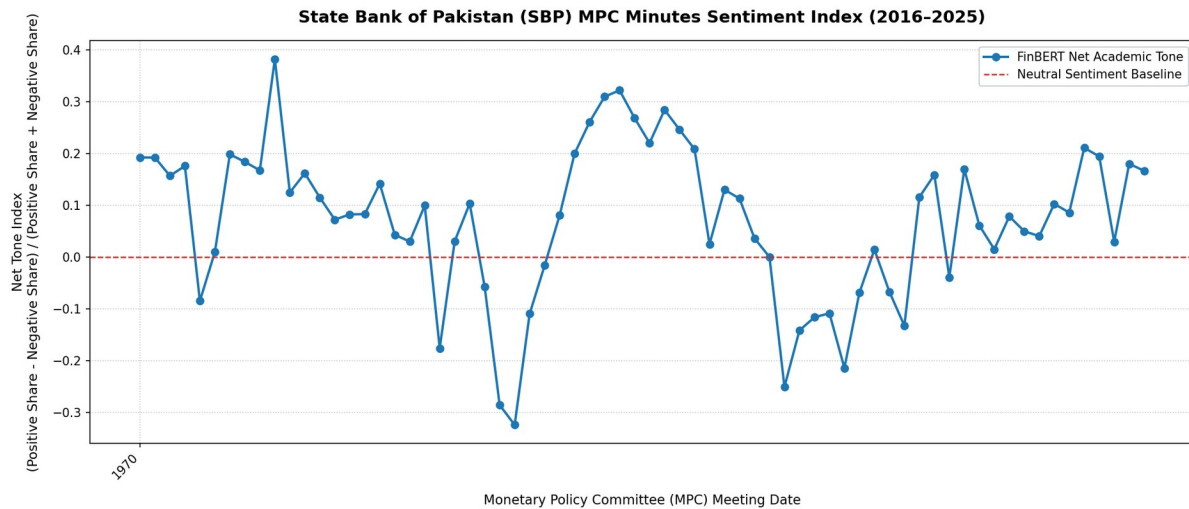


Figure 1. FinBERT net tone index for SBP MPC minutes from 2016 to 2025.

The tone series shows variation across MPC meetings, rather than staying flat around zero. This is important because the tone index can then be treated as a measurable communication variable in a market reaction model.

3. Market Matching and Regression Setup

After constructing the tone index, the project matches each MPC release date with the next available KSE-100 trading date. This avoids matching policy communication to a non-trading day and makes the market reaction window more realistic.

The baseline regressions use KSE-100 log returns as the dependent variable. Predictors include current and lagged FinBERT tone, lagged KSE returns, current and lagged KIBOR, and differenced variables in the error-correction form.

Lag selection was performed by looping over ARDL-style combinations: 1 to 4 lags of returns, 1 to 4 lags of tone, and 1 to 4 lags of KIBOR. The lowest-AIC specification from the search was ARDL(2,1,1).

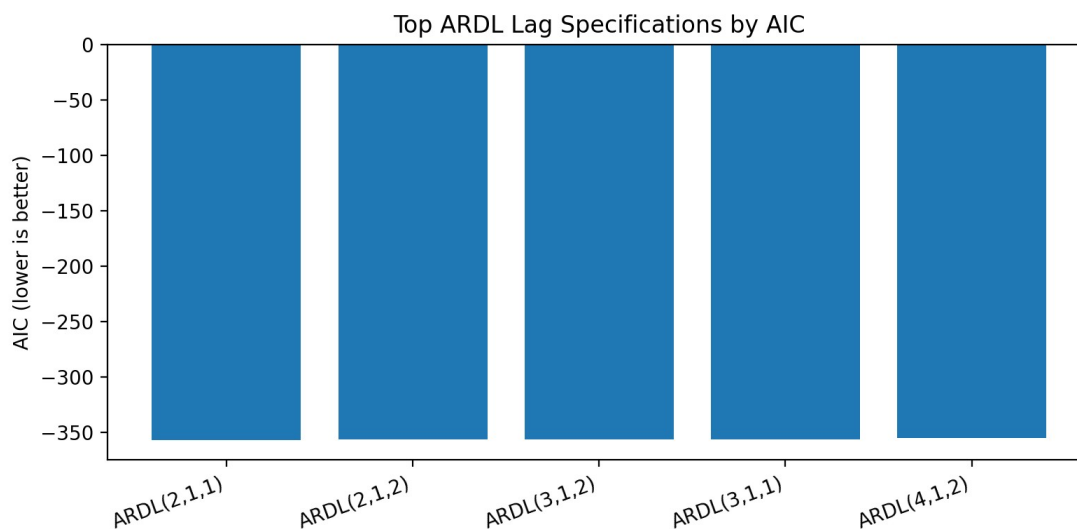


Figure 2. Best lag specifications ranked by AIC. Lower AIC is preferred.

Rank	p	q1	q2	AIC
1	2	1	1	-356.819
2	2	1	2	-356.652
3	3	1	2	-356.544

4. Main Econometric Results

A direct ARDL-style OLS model using returns, tone, and KIBOR lags produced limited explanatory power. The model had R-squared of 0.161 and adjusted R-squared of 0.044. In that model, the second lag of returns was statistically significant, but current tone and lagged tone were not statistically significant.

The model was then re-expressed as an unrestricted error correction model. This version had stronger fit, with R-squared of 0.537 and adjusted R-squared of 0.488. The overall F-test was significant, suggesting the specification explains variation in differenced returns better than the simple static approach.

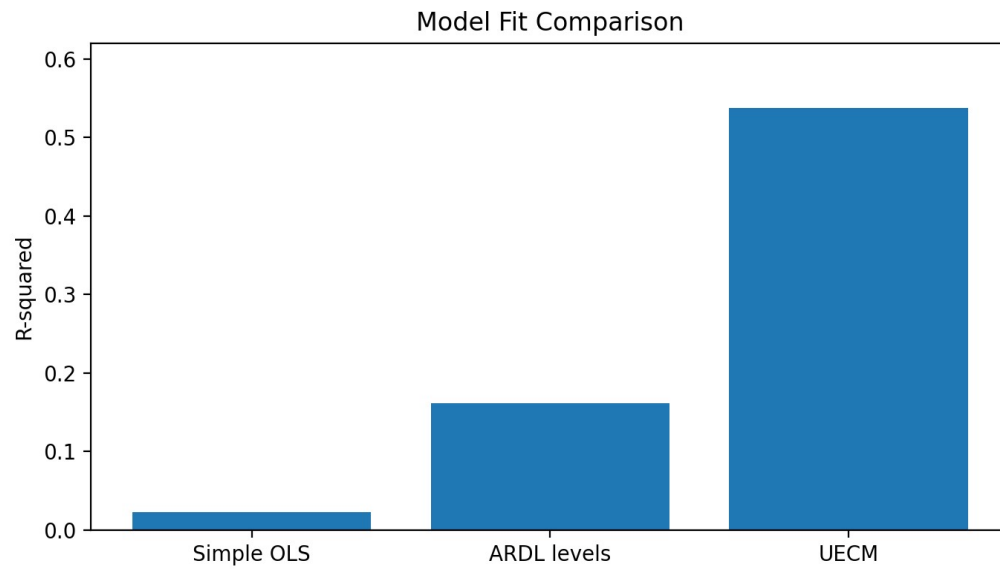


Figure 3. Comparison of model fit across simple OLS, ARDL-style levels model, and UECM.

Model	R-squared	Adj. R-squared	Interpretation
Simple OLS	0.023	-0.009	Very weak contemporaneous relationship.
ARDL-style levels model	0.161	0.044	Some fit, mainly driven by lagged returns.
UECM	0.537	0.488	Best fit among tested models.

5. UECM and Cointegration Evidence

The UECM was used to test whether returns, FinBERT tone, and KIBOR share a long-run relationship. The initial bounds-style Wald F-test on lagged levels produced $F = 17.29$ with $p < 0.001$, which suggests cointegration under conventional interpretation.

However, after using HC3 robust standard errors because heteroskedasticity was detected, the same joint test weakened to $F = 2.67$ with $p = 0.056$. This means the long-run relationship should be treated as suggestive rather than conclusive.

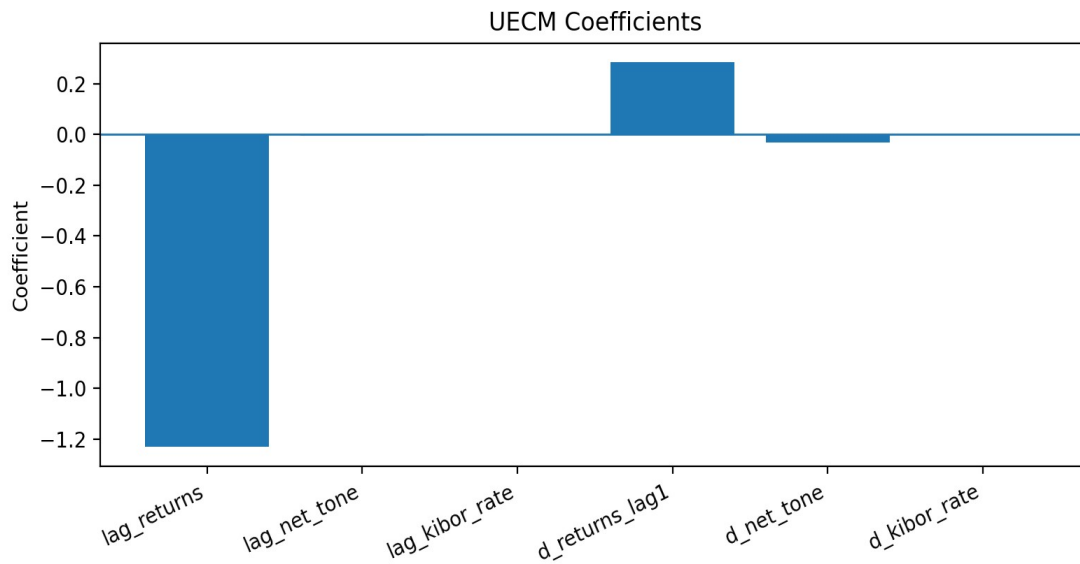


Figure 4. UECM coefficient estimates for lagged levels and differenced variables.

Variable	Coef.	p-value	Interpretation
lag_returns	-1.2319	0.000	Strong correction from previous return level.
lag_net_tone	-0.0065	0.713	Lagged tone not statistically significant.
d_log_returns_lag1	0.2841	0.030	Short-run return momentum effect.
d_net_tone	-0.0305	0.119	Tone change is not significant at 5%.

6. Diagnostics and Robustness

The Breusch-Godfrey serial correlation test did not detect residual autocorrelation. The LM p-value was 0.466 and the F-test p-value was 0.540, both comfortably above 0.05.

The Breusch-Pagan test did detect heteroskedasticity. The LM p-value was 0.007 and the F-test p-value was 0.004. Because of this, HC3 robust standard errors were used as a robustness check.

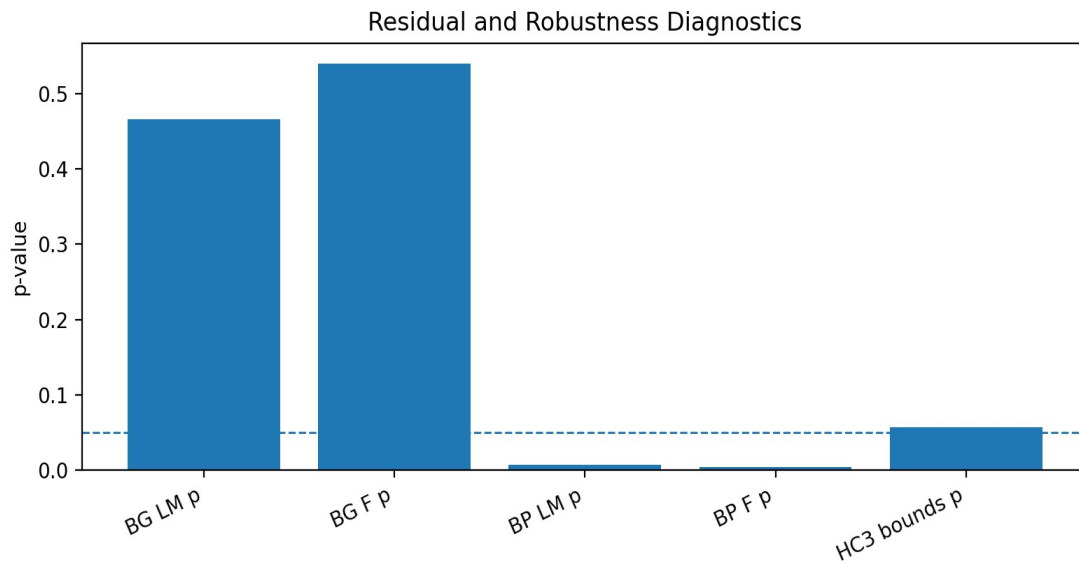


Figure 5. Diagnostic p -values. The 0.05 line marks conventional statistical significance.

7. Simple OLS Benchmark

As a simple benchmark, KSE-100 log returns were regressed on current FinBERT net tone and the change in KIBOR. This model had R-squared of only 0.023 and an overall p-value of 0.497.

The tone coefficient was -0.0111 with $p = 0.421$, which means the direct contemporaneous relationship between policy tone and same-window market returns was not statistically significant in this specification.

Variable	Coef.	p-value	Meaning
net_tone	-0.0111	0.421	Not statistically significant.
d_kibor_rate	-0.0013	0.411	Not statistically significant.
Model R-squared	0.023	0.497 overall	Very limited explanatory power.

8. Limitations and Future Work

- The project uses FinBERT on SBP MPC minutes, but the model is trained mainly on general financial text, not specifically Pakistani central-bank communication.
- The sample size is small for time-series econometrics because MPC meetings occur only a few times per year.
- The market reaction window is simple; future work can test 1-day, 3-day, and 5-day cumulative abnormal returns.
- The model controls for KIBOR, but a fuller specification should include inflation surprises, exchange rate changes, policy rate surprises, and global risk variables.
- The robust bounds test weakens the cointegration result, so the long-run relationship should not be overstated.
- Future work can compare FinBERT with domain-specific central bank dictionaries, Loughran-McDonald sentiment, and manually validated SBP tone labels.

9. Conclusion

This project converts qualitative SBP monetary policy communication into a measurable financial tone index using FinBERT, then connects that tone index to KSE-100 market returns. The strongest contribution is the full pipeline: text extraction, sentence-level financial sentiment scoring, document-level tone aggregation, market-date matching, lag selection, ARDL/UECM modeling, and diagnostic testing.

The empirical result is careful rather than exaggerated. FinBERT tone appears useful as a structured communication variable, but in this sample it does not independently explain short-run KSE-100 returns in a statistically strong way. A stronger future version should add event-study abnormal returns, policy surprise measures, and richer macro-financial controls.